

DISCRIMINATION IN THE PRESENCE OF COVARIABLES

Kathleen Subrahmaniam
University of Manitoba, Canada

SUMMARY

In this paper we develop a unified approach to the theory of discriminant analysis in the presence of covariables by partitioning the D^2 -statistic. We demonstrate the equivalence of the methods of Rao (1949) and Cochran and Bliss (1948). A study is made of the null and non-null distributions of the test statistic, which is a function of D^2 , for various hypotheses concerning the equality of mean vectors.

1. Introduction

Suppose we wish to discriminate between two populations when the measurement vector $\mathbf{Y}$ is composed of both discriminators and covariables. We shall assume that the populations differ only with respect to these discriminators or main variables. Although the covariables (or concomitant variables) do not discriminate between populations, they may provide an improvement in the discriminating power of the main variables.

Frequently in the literature the term, "discriminant analysis", is used to refer to classificatory problems; however, in this paper we shall use the discriminant analysis as a tool for testing the equality of mean vectors in two populations.

To fix our ideas, let us consider a $p + q$ vector of measurements $\mathbf{Y}' = (\mathbf{X}', \mathbf{Z}')$, where $\mathbf{X}$ is a p -dimensional vector and $\mathbf{Z}$ is q -dimensional. We shall refer to $\mathbf{X}$ as the discriminators and $\mathbf{Z}$ as the covariables. Our problem then reduces to one of testing whether the expected value in the two populations is the same, where $\mathbf{Y}_i \approx N_{p+q}(\mu_i, \Sigma)$ for $i = 1, 2$.

Expanding on the ideas of Rao (1966) we shall consider several hypotheses which might be of interest:

- (a) $E(\mathbf{Y}_1) = E(\mathbf{Y}_2)$
- (b) $E(\mathbf{X}_1 | \mathbf{Z}_1) = E(\mathbf{X}_2 | \mathbf{Z}_2)$
- (c) $E(\mathbf{X}_1 | \mathbf{Z}_1) = E(\mathbf{X}_2 | \mathbf{Z}_2)$ given $E(\mathbf{Z}_1) = E(\mathbf{Z}_2)$
- (d) $E(\mathbf{X}_1) = E(\mathbf{X}_2)$ given $E(\mathbf{Z}_1) = E(\mathbf{Z}_2)$
- (e) $E(\mathbf{X}_1) = E(\mathbf{X}_2)$

Let us now consider the circumstances under which each of these tests might be appropriate. Hypothesis (a) is a test of the equality of the mean vector including both main variables and covariables. In this situation we would be assuming that the covariables can be used to discriminate between the two populations and that they are not merely auxiliary variables to be used to increase the precision of the test.

In hypothesis (b) we are interested in a conditional hypothesis. This hypothesis is a test of the equality of the mean vectors of the main variables adjusted for the effect of the covariables. Here, as in the univariate covariance analysis, we are hoping to increase the precision of the test by adjusting for the effect of the covariables.

Hypothesis (c) is a restricted form of the conditional hypothesis in (b). Here we are assuming the equality of the mean vectors of the covariables and that the multiple regression of $\bar{X}$ on $\bar{Z}$ is the same in both populations. We shall see that the null distributions of the test statistics in (b) and (c) are identical.

Although hypothesis (d) can also be shown to be equivalent to (c), it is stated in a more general form. As it is an unconditional hypothesis, we are no longer considering the covariables as fixed.

In most situations the assumption that the populations do not differ in respect to their covariables seems realistic. If in fact they do differ, the covariables should probably be included as main variables rather than as concomitant ones.

If we find that the correlations between main variables and covariables are small, it is clear that the covariables will not provide an increase in the precision. In this case we may wish to ignore the covariables and test hypothesis (e).

Our main interest will be hypotheses (b), (c) and (d). Although tests of these hypotheses have been developed by Rao (1948, 1949, 1952, 1966) and Cochran and Bliss (1948), it is our aim to present a unified approach to these tests by partitioning the Mahalanobis's D^2_{p+q} (1925, 1930) statistic under various conditions. This partitioning allows for more direct and logical development of the distributions of these statistics. Using this technique, null and non-null distributions are obtained for each of the test statistics corresponding to the five hypotheses given above.

2. Notation and Partitioning of the Distance Function (D^2)

Let $(y_{i1}, y_{i2}, \dots, y_{in_i})$ for $i = 1, 2$ be a sample of n_i observations for a $(p+q)$ variate normal distribution $N_{p+q}(\mu_i, \Sigma)$. The sample difference vector $\underline{d} = \bar{y}_1 - \bar{y}_2$ is then an estimator for $\delta = \mu_1 - \mu_2$ and S , the pooled sample dispersion matrix based on $n_1 + n_2 - 2$ degrees of freedom, is an estimator for Σ . In what follows we shall denote $n_1 + n_2$ by N and $n_1 n_2 / (n_1 + n_2)$ by c .

Let us now consider the distance function $D^2_{p+q} = \underline{d}' S^{-1} \underline{d}$, where $p+q$ designates the number of characteristics on which the statistic is based.

We know that

$$(1) \quad \frac{(N-p-q-1) c D^2_{p+q}}{(p+q)(N-2)} \approx F^*(p+q, N-p-q-1, \underline{\delta}' S^{-1} \underline{\delta}),$$

where $F^*(\nu_1, \nu_2, \lambda^2)$ represents a noncentral F -distribution with ν_1 and ν_2 degrees of freedom and noncentrality parameter λ^2 . Under the null hypothesis ($\underline{\delta} = \underline{0}$), the statistic in (1) is distributed $F(p+q, N-p-q-1)$, where $F(\nu_1, \nu_2)$ is a central F -distribution with ν_1 and ν_2 degrees of freedom.

Using these results, hypothesis (a) and (e) can be tested. Note that for (e) $q = 0$.

In the remaining development we shall find the following notation useful:

Let $\underline{d}$ and S be partitioned into components corresponding to p variables and q covariables. The distance function of the discriminators adjusted for the covariables can then be symbolized as

$$(2) \quad D^2_{p|q} = (\underline{d}_1 - S_{12} S_{22}^{-1} \underline{d}_2)' [S_{11} - S_{12} S_{22}^{-1} S_{21}]^{-1} (\underline{d}_1 - S_{12} S_{22}^{-1} \underline{d}_2)$$

and the distance function of the covariables as

$$(3) \quad D_q^2 = \underline{d}_2' S_{22}^{-1} \underline{d}_2.$$

A similar notation can be used for the population distance functions $\Delta_{p|q}^2$ and Δ_q^2 , which are functions of $\underline{\delta}$ and Σ .

Using (2) and (3) and appealing to a familiar theorem in quadratic forms, we can show that

$$(4) \quad D_{p+q}^2 = D_{p|q}^2 + D_q^2,$$

that is, the sample D_{p+q}^2 statistic based on p variables and q covariables can be partitioned into the sum of the sample distance function for the p variables conditioned by the q covariables and the distance function for the q covariables.

We shall use (4) as a basis for a unified approach for developing tests for the hypotheses suggested in section (1).

3. Test of the Hypothesis: $E(\underline{X}_1 | \underline{Z}_1) = E(\underline{X}_2 | \underline{Z}_2)$

Using the results of section 2, we shall develop tests involving the equality of the conditional expectations in the two populations. The hypothesis

$$E(\underline{X}_1 | \underline{Z}_1) = E(\underline{X}_2 | \underline{Z}_2)$$

can be stated alternatively as

$$\Delta_{p|q}^2 = \Delta_{p+q}^2 - \Delta_q^2 = 0.$$

This is a test of whether or not the distance function for the main variables adjusted for the covariables is different from zero. Note that here no assumption is made concerning the expectation of the covariables.

Recall that the vector variables $\underline{Y}_i'$ for $i = 1, 2$ can be partitioned as

$$\underline{Y}_i = \begin{bmatrix} \underline{X}_i \\ \underline{Z}_i \end{bmatrix} \approx N_{p+q} \left(\begin{bmatrix} \underline{\mu}_{X_i} \\ \underline{\mu}_{Z_i} \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix} \right).$$

In each population let us transform these original variables $\underline{Y}$ into two components $\{\underline{U}_1, \underline{U}_2\}$ such that $\underline{U}_1$ with p components and $\underline{U}_2$ with q components are uncorrelated. This can be achieved by the transformation

$$\underline{U} = T \underline{Y},$$

where

$$T = \begin{bmatrix} I & -\Sigma_{12} \Sigma_{22}^{-1} \\ 0 & I \end{bmatrix}$$

We then know that

$$\underline{U}_{1i} \approx N_p(\underline{\mu}_{X_i} - \Sigma_{12} \Sigma_{22}^{-1} \underline{\mu}_{Z_i}, \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})$$

$$\underline{U}_{2i} \approx N_q(\underline{\mu}_{Z_1}, \Sigma_{22}).$$

From (2) we can write $D_{p|q}^2 = \underline{d}'^* S^{*-1} \underline{d}^*$, where $\underline{d}^* = \underline{\bar{U}}_{11}' - \underline{\bar{U}}_{12}' \underline{\bar{U}}_{22}^{-1} \underline{S}_{21}$. Obtaining the distribution of $D_{p|q}^2$ is facilitated by first finding the distribution of $\underline{d}^*$ and S^* . It can be shown that

$$\underline{d}^* \approx N_p[\underline{\delta}_1 - \Sigma_{12} \Sigma_{22}^{-1} \underline{\delta}_2, \nu(\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})]$$

where

$$\nu = \frac{1}{c} + \frac{1}{N-2} D_q^2$$

and $(N-2-q)S^*$ is distributed as Wishart (W_p) with parameters $N-2-q$ and $\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$. Since $\underline{d}^*$ and S^* are independent, it follows, from a development similar to that leading to (1), that

$$(5) \quad \frac{N-p-q-1}{p\nu(N-2)} D_{p|q}^2 \approx F(p, N-p-q-1, \tau^2)$$

where

$$\tau^2 = \frac{1}{\nu} [\underline{\delta}_1' (\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})^{-1} \underline{\delta}_1]$$

with

$$\underline{\delta}_1^* = [\underline{\mu}_{X_1} - \underline{\mu}_{X_2} - \Sigma_{12} \Sigma_{22}^{-1} (\underline{\mu}_{Z_1} - \underline{\mu}_{Z_2})]$$

An analogous test criterion has been derived by Rao (1949). Rao develops the test from a somewhat different point of view. He considers the contribution of the covariables to the distance to be fixed at D_q^2 and then he is interested in finding whether or not there has been an increase in the distance due to the main variables. Essentially he finds the distribution of

$$R = \frac{1 + \frac{c}{N-2} D_q^2}{1 + \frac{c}{N-2} D_q^2 + p}$$

for a fixed D_q^2 . He finds the conditional distribution of R given D_q^2 to be type I beta with parameters $(N-p-q-1)/2$ and $p/2$ and noncentrality parameter

$$\lambda^2 = \frac{c \underline{\delta}_1' (\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})^{-1} \underline{\delta}_1}{1 + \frac{c}{N-2} D_q^2}$$

It can easily be seen that the distribution of $(1-R)/R$ is identical with that in (5).

At this point it may be helpful to discuss more fully the implications of the hypotheses under test. In the terminology that has been developed in this section, we are interested in testing the hypotheses

$$E(\underline{X}_1 | \underline{Z}_1) = E(\underline{X}_2 | \underline{Z}_2),$$

which is equivalent to

$$(6) \quad \underline{\delta}_1^* = (\underline{\mu}_{X_1} - \underline{\mu}_{X_2}) - \Sigma_{12} \Sigma_{22}^{-1} (\underline{\mu}_{Z_1} - \underline{\mu}_{Z_2}) = \underline{0}.$$

We have shown that the test can be based on the statistic $D_{p|q}^2$. Analogous to the independent variables in the multiple regression situation, we have assumed the covariables to be fixed and are considering only a conditional hypothesis.

Under the null hypothesis the statistic

$$\frac{N-p-q-1}{p(N-2)} \frac{c}{\nu} D_{p|q}^2 \approx F(p, N-p-q-1).$$

Note that although ν is a function of D_q^2 , it is a constant since we are considering only a conditional hypothesis.

4. Test of the Hypothesis: $E(\underline{X}_1 | \underline{Z}_1) = E(\underline{X}_2 | \underline{Z}_2)$ Given that $E(\underline{Z}_1) = E(\underline{Z}_2)$

Although Cochran and Bliss (1948) were essentially interested in this same problem, namely, adjusting the distance function of the main variables for the effect of the covariables, their development parallels that of a multiple regression technique. In their procedure the multiple regression is obtained for each of the main variables on all the covariables. Each main variable is then replaced by its deviations from multiple regression and a new distance function is calculated in the usual way using these deviations. Cochran and Bliss make a further assumption that the covariables have the same population means, that is $\underline{\mu}_{Z_1} = \underline{\mu}_{Z_2}$. Imposing this restriction, the hypothesis under test becomes

$$E(\underline{X}_1 | \underline{Z}_1) = E(\underline{X}_2 | \underline{Z}_2) \text{ given that } E(\underline{Z}_1) = E(\underline{Z}_2)$$

or

$$\underline{\delta}_{1R}^* = \underline{\mu}_{X_1} - \underline{\mu}_{X_2} = \underline{0},$$

where the subscript R designates that the general hypothesis in (6) has been restricted and is no longer a function of the expected values of the covariables.

We can obtain a test for this restricted hypothesis by setting $\underline{\mu}_{Z_1} - \underline{\mu}_{Z_2}$ equal to zero in the noncentrality parameter of (5). The conditional distribution of $D_{p|q}^2$ for a given D_q^2 then becomes

$$(7) \quad \frac{N-p-q-1}{p \nu (N-2)} D_{p|q}^2 \approx F(p, N-p-q-1, \tau_R^2),$$

where

$$\tau_R^2 = \frac{1}{\nu} [\underline{\delta}_{1R}^{*'} (\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})^{-1} \underline{\delta}_{1R}^*]$$

The distribution given in (7) is equivalent to that obtained by multiple regression techniques by Cochran and Bliss.

The null distributions in (5) and (7) are equivalent; however, the power functions will be different since the noncentrality parameter in the restricted case is not equivalent to that in the unrestricted case. It is the view of the author that this restricted hypothesis may in practice be more applicable than the unrestricted one.

5. Test of the Hypothesis: $E(\underline{X}_1) = E(\underline{X}_2)$ Given that $E(\underline{Z}_1) = E(\underline{Z}_2)$

In the previous sections we were interested in the conditional distribution of $D_{p|q}^2$ for a fixed D_q^2 . We are now interested in testing $E(\underline{X}_1) = E(\underline{X}_2)$ under the assumption that $E(\underline{Z}_1) = E(\underline{Z}_2)$. This hypothesis can be restated as $\delta_1 = \underline{\mu}_{X_1} - \underline{\mu}_{X_2} = \underline{0}$ when $\underline{\mu}_{Z_1} = \underline{\mu}_{Z_2}$.

It is immediately obvious that this hypothesis is equivalent to that in section 4; however, we shall develop an alternative test which allows for variation in D_q^2 . The statement of the hypothesis suggests that we are interested in the unconditional distribution of $D_{p|q}^2$. At this point it may be helpful to recall that $D_{p|q}^2$ can be written as $D_{p+q}^2 - D_q^2$. If we develop the unconditional distribution of this difference, we can allow for variation in D_q^2 . We will no longer be treating the covariables as fixed as we did in sections 3 and 4.

As we already know the distribution of $D_{p+q}^2 - D_q^2$ for a fixed D_q^2 , let us now develop the distribution of $D_{p+q}^2 - D_q^2$. Using similar techniques to those of section 3, we can show that

$$(8) \quad \frac{N-q-1}{q(N-2)} c D_q^2 \approx F(q, N-q-1, c \tau^2)$$

where

$$\tau^2 = \delta_1' \Sigma_{22}^{-1} \delta_1$$

We shall first consider the unconditional distribution of $(D_{p+q}^2 - D_q^2)$ under the null hypothesis; hence $\delta_1 = \underline{0}$ and $\delta_2 = \underline{\mu}_{Z_1} - \underline{\mu}_{Z_2} = \underline{0}$, which implies that the noncentrality parameters in (7) and (8) are both zero.

In order to find the unconditional distribution, let us consider the transformation

$$U = \frac{c}{N-2} D_q^2$$

and

$$(9) \quad X = \frac{c(D_{p+q}^2 - D_q^2)}{N-2 + c D_q^2}$$

It can be shown that if $Y \approx F(r, s)$, then $\frac{r}{s} Y \approx B'(r, s)$, where $B'(r, s)$ represents a type II beta distribution (Rao (1965, p. 145)). Applying this result to (7) and (8), we can write the conditional distribution of X for a given U and the marginal of U under the null hypothesis as

$$g_1(x|u) = B'(x; p, N-p-q-1)$$

and

$$g_2(u) = B'(u; q, N-q-1).$$

Since we shall be interested in the marginal distribution of

$$(10) \quad T = \frac{c}{N-2} (D_{p+q}^2 - D_q^2),$$

let us make the transformation $T = X(1+U)$ and $V = U$. The joint distribution of the transformed variables will then be

$$g(t,v) = C \frac{t^{(p/2)-1} v^{(q/2)-1}}{(1+v)^{(p+q)/2} (1+t+v)^{(N-q-1)/2}} \quad 0 < t, v < \infty,$$

where

$$C = \frac{\Gamma(\frac{N-q-1}{2}) \Gamma(\frac{N-1}{2})}{\Gamma(\frac{p}{2}) \Gamma(\frac{q}{2}) \Gamma(\frac{N-p-q-1}{2}) \Gamma(\frac{N-q-1}{2})}.$$

Writing $g(t,v)$ in terms of an infinite series and noting that the integral

$$\int_0^\infty \frac{v^{(q/2)+r-1}}{[1+v]^{[(N+p-1)/2]+r}} dv$$

is the beta function

$$\frac{\Gamma(\frac{q}{2} + r) \Gamma(\frac{N+p-q-1}{2})}{\Gamma(\frac{N+p-1}{2} + r)},$$

we can then write the marginal distribution of T as

$$h_1(t) = C \frac{t^{(p/2)-1}}{[1+t]^{(N-q-1)/2}} \sum_{r=0}^{\infty} \left[\frac{t}{1+t} \right]^r \frac{\Gamma(\frac{N-q-1}{2} + r) \Gamma(\frac{q}{2} + r) \Gamma(\frac{N+p-q-1}{2})}{\Gamma(\frac{N-q-1}{2})! \Gamma(\frac{N+p-1}{2} + r)}$$

Recalling that the hypergeometric function of the second kind is defined as (Erdélyi, et al. (1953), Chapter 5)

$${}_2F_1(a, b; c; x) = \sum_{n=0}^{\infty} \frac{\Gamma(a+n) \Gamma(b+n) \Gamma(c)}{\Gamma(a) \Gamma(b) \Gamma(c+n)} \frac{x^n}{n!},$$

$h_1(t)$ can be written as

$$(11) \quad h_1(t) = \frac{\Gamma(\frac{N+p-q-1}{2}) \Gamma(\frac{N-1}{2})}{\Gamma(\frac{N-p-q-1}{2}) \Gamma(\frac{p}{2}) \Gamma(\frac{N+p-1}{2})} \frac{t^{(p/2)-1}}{[1+t]^{(N-q-1)/2}} {}_2F_1\left(\frac{q}{2}, \frac{N-q-1}{2}, \frac{N+p-1}{2}; \frac{t}{1+t}\right)$$

This same result was obtained by Rao (1949) by quite a different argument.

The percentage points (0.90, 0.95, 0.99) for the distribution in (11) have been tabulated by Subrahmaniam and Subrahmaniam (1970). Rao (1949, 1950) has suggested two approximations to (11):

$$\frac{(N-q-1)(N-p-q-1)}{(N-1)p} T \approx F(p, N-p-q-1)$$

and

$$\frac{(N-q-1)(N-p-q-1)}{(N-1)} T \approx \chi^2(p).$$

He indicated that the latter would be superior for small p . Subrahmaniam and Subrahmaniam (1970) have compared these two approximations with the exact distribution given in (11).

Let us now consider the non-null distribution of T as defined in (10), that is, the distribution of T when $\underline{\delta}_1 \neq \underline{0}$ and $\underline{\delta}_2 = \underline{0}$. Using the same transformation as in the central case, we see that now

$$g_1^*(x|u) = B' \left(x; p, N-p-q-1, \frac{1}{\nu} [\underline{\delta}'_1 (\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})]^{-1} \underline{\delta}_1 \right)$$

and

$$g_2^*(u) = B'(u; q, N-q-1),$$

where $B'(r, s, \lambda^2)$ is a noncentral type II beta distribution (Rao (1965, p. 175)). The corresponding joint distribution then becomes

$$\begin{aligned} f^*(x, u) &= \frac{\Gamma(\frac{N-1}{2})}{\Gamma(\frac{q}{2})\Gamma(\frac{N-q-1}{2})} \frac{u^{(q/2)-1}}{[1+u]^{(N-1)/2}} \\ &\sum_{r=0}^{\infty} \frac{(ca^2/2)\exp[-ca^2/2(1+u)]\Gamma(\frac{N-q-1}{2}+r)}{r! (1+u)^r \Gamma(\frac{p}{2}+r)\Gamma(\frac{N-p-q-1}{2})} \frac{x^{(p/2)+r-1}}{[1+x]^{(N-q-1)/2+r}}, \end{aligned}$$

where

$$a^2 = \underline{\delta}'_1 (\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21})^{-1} \underline{\delta}_1.$$

Let us again consider the transformation used previously, that is, $T = X(1+U)$ and $V = U$. We then see that

$$g^*(t, v) =$$

$$\sum_{r=0}^{\infty} \sum_{s=0}^{\infty} K_{rs} \frac{v^{(q/2)-1}}{[1+v]^{(N-1)/2+r+s+1}} \frac{t^{(p/2)-1}}{[1+v]^{(p/2)+r-1}} \frac{[1+v]^{(N-q-1)/2+r}}{[1+t+v]^{(N-q-1)/2+r}}$$

where

$$K_{rs} = \frac{(-1)^s \Gamma(\frac{N-1}{2}) \Gamma(\frac{N-q-1+r}{2})}{s! r! \Gamma(\frac{q}{2}) \Gamma(\frac{N-q-1}{2}) \Gamma(\frac{p+r}{2}) \Gamma(\frac{N-p-q-1}{2})} (ca^2/2)^{r+s}$$

Following the development of the null distribution, the marginal distribution of T is obtained as

$$h_1^*(t) = \sum_{r=0}^{\infty} \sum_{s=0}^{\infty} K_{rs} \frac{t^{(p/2)+r-1}}{[1+t]^{\{(N-q-1)/2\}+r}} \sum_{j=0}^{\infty} \frac{\Gamma(\frac{N-q-1}{2}+r+j) \Gamma(\frac{q+j}{2})}{\Gamma(\frac{N-q-1}{2}+r)} \frac{\Gamma(\frac{N+p-q-1}{2}+2r+s)}{\Gamma(\frac{N+p-1}{2}+2r+s+j)} \frac{1}{j!} \left[\frac{1}{1+t} \right]^j$$

Replacing K_{rs} by the corresponding gamma functions and recalling the definition of the hypergeometric function of the second kind, we find that

$$(12) \quad h_1^*(t) = \sum_{r=0}^{\infty} \sum_{s=0}^{\infty} \frac{(-1)^s}{r! s!} \left\{ \frac{ca^2}{2} \right\}^{r+s} \frac{\Gamma(\frac{N-1}{2}) \Gamma(\frac{N-q-1}{2}+r)}{\Gamma(\frac{p+r}{2}) \Gamma(\frac{N-q-1}{2}) \Gamma(\frac{N-p-q-1}{2})} \frac{\Gamma(\frac{N+p-q-1}{2}+2r+s)}{\Gamma(\frac{N+p-1}{2}+2r+s)} \frac{t^{(p/2)+r-1}}{[1+t]^{\{(N-q-1)/2\}+r}} {}_2F_1\left(\frac{q}{2}, \frac{N-q-1}{2}+r; \frac{N+p-1}{2}+2r+s; \frac{t}{1+t}\right).$$

A similar result has been indicated by Rao (1949) although he does not actually obtain the marginal distribution of T.

Having obtained the non-null distributions of the test statistics corresponding to hypotheses (c) and (d), we can compare the power of the test based on the conditional ratio with that of the test based on the unconditional difference when $\delta_2 = 0$. Using (7) and (12) the power of these two tests has been computed for various p, q and N by Subrahmaniam and Subrahmaniam (1970).

Rao (1950) has developed a noncentral distribution. Close inspection will reveal that it is not the same as (12). Rather it is the distribution of T when $\delta_1 \neq 0$ and $\delta_2 \neq 0$. It is the non-null distribution of the test statistic for an hypothesis of the form $E(\bar{X}_1) = E(\bar{X}_2)$ when no assumption is made concerning the expectation of the covariables. The author, however, feels that hypothesis (d) is more realistic. If no assumption is made concerning the covariables, perhaps hypothesis (a) or (e) is more fitting. Since the noncentral distribution obtained by Rao has not been tabulated, the power of these various tests cannot be compared.

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